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Motion Prejudgment Dependent Mixture System Noise in System Model for Tennis Ball 3D Position Tracking by Particle Filter

Yuan Wang[1], Xina Cheng[1], Norikazu Ikoma[2], Masaaki Honda[3] , Takeshi Ikenaga[1]
[1]Graduate School of Information, Production and System, Waseda University, Kita-Kyushu, Fukuoka, Japan
[2] Dept. of Computer and Information Engineering, Faculty of Engineering, Nippon Institute of Technology, Saitama, Japan.
[3] Faculty of Sport Sciences, Waseda University, Tokyo, Japan
Email: wangyuan@suou.waseda.jp

***Abstract*—In tennis game analysis, the 3D position of ball plays a crucial role in score judgment and player evaluation. When tracking the tennis ball in 3D space, high speed and abrupt motion change of the tennis ball are the main problems which make it difficult to predict the near future course of the ball. Aiming at solving above two problems, we propose a system model based on an elaborated mixture system noise. The mixture system noise consists of general change noise and adaptive abrupt change noise which is dependent on motion prejudgment result of tennis ball. The motion prejudgment method is carried out by the current state of ball and players. The motion of ball is classified into general motion and three abrupt motions, including smash, bounce and hit the net. Experiments based on 13 HDTV video sequences, which were recorded by four cameras located at four corners of the tennis court outside in a cloudy day including two players were used to explore the performance of the proposed method. The tracking success rate is 81.14%, gaining 27.64% improvement compared with conventional work.**

Keywords—tennis ball tracking; mixture system noise; motion prejudgment; particle filtering

I. INTRODUCTION

Tracking of tennis ball position in 3D space is of prime importance in tennis game analysis since it is the key element to game score judgment and player ability evaluation. In score judgment application, the real 3D position of the ball is required in order to compare it with the reference object position. In game technical essential analysis and player evaluation aspect, the position and velocity of ball can be used to indicate the player's skill and power which reveal the advantage and disadvantage. Therefore, the tennis ball tracking method attracts an increasing attention over these years. Our research targets improving the success rate of tracking the 3D position of tennis ball for tennis game analysis.

There have been three unique problems in tracking the 3D position of tennis ball, comparing with other ball sports, which influence the tracking success rate significantly. The first one is the high speed and abrupt motion change of ball, which leads to a big position difference between two consecutive states step. In other words, it is difficult to give an appropriate prediction to the near future course of the target when the motion of it changes abruptly and quickly in tracking process. Secondly, the precise 3D coordination is hard to obtain. Thirdly, the occlusion problem caused by other objects and the limited location of cameras have a bad impact on getting stable tracking result of the target. Besides these unique problems, there remain some common influence factors, such as the small size of ball, non-ideal camera parameters and video shooting environment including cloudy and rainy weather.

Several methods aiming at tracking tennis ball have been proposed over these years and some of them are summarized in the following. In Yan's work [1], based on a low-quality off-air video recorded by a single camera, they use particle filter to track tennis ball candidates which are explored by multiple visual cues and propagate particles by a switch of two dynamic models. The Zhou's [2] proposal is aiming at improving the robustness of tennis ball tracking with a two-layered data association method to find out a unique ball trajectory from a set of short trajectories. However, these two works have little to do with the 3D position of tennis ball. Hawk-Eye system [3] has currently been applied widely for numerous sports to track the trajectory of the ball and display it for score judgment. When it refers to the Hawk-eye system designed for tennis game, the system uses ten high-speed and high-performance cameras which normally placed at a high position from different angles to get the visual images and timing data. Then after computation, it displays the most likely path of the moving ball. The limitation of this system is that it requires large number, high-quality and well-arranged place location of the cameras. On account of the tennis game is mostly held outside, the changing weather leads to big difference in features captured from the target tennis ball.

Cheng's work [4] has put forward a multi-view ball tracking method by particle filter for tracking ball position in 3D space for volleyball analysis. This work has proposed a 3D framework of particle filter [5] with ball feature likelihood for 3D ball tracking in volleyball analysis and can avoid the error produced by 3D structure reconstruction [6]. Nonetheless, that work is weak at dealing with the target which has high speed and abrupt motion change since its system model for predicting position of ball in next state step cannot cover all the situations.

At the same time, there have been many methods proposed to track different targets by particle filter. Guo [7] has enlarged the search region based on the algorithm of particle filter in order to solve the temporally occlusion problems for tracking human. Ikoma [8] has added location factor and shape factor into the hidden state of the object and done motion estimation to track the deformable target. Hess [9] has put forward a discriminatively trained particle filter with the HSV likelihood model to make color information stable.

To tackle with the high speed and abrupt motion change problems of target in tennis ball tracking, we propose further improvement of system model along with the framework of multi-view ball tracking method by particle filter which is built by Cheng's work [4]. The system model is designed by time

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evolution of former state vector and motion prejudgment dependent mixture system noise to get tracking candidates of target in next state step. The mixture system noise consists of general change noise and adaptive abrupt change noise. Mixture ratio and certain abrupt motion are decided by the motion prejudgment result. The motion prejudgment method is executed by current state of ball and players based on the definition of tennis ball motion. The motion of ball is classified into general motion and three abrupt motions, smash, bounce and hit the net. The state vector consists of position and velocity in 3D space of tennis ball. Experiment videos were taken outside by four cameras located at the four corners of tennis court in a cloudy day.

This paper is arranged as follows. Section Ⅱ and section Ⅲ covers the detail of multi-view ball tracking method and our proposals. Then next section Ⅳ is the experiment and discussion. Finally, conclusion is in section Ⅴ.

II. MULTI-VIEW BALL TRACKING METHOD

Overall of the 3D ball tracking framework, multi-view ball tracking method by particle filter with our proposals is introduced. The whole structure is shown in Fig.1. It can solve the one direction occlusion problem and obtain 3D position and velocity of target ball. We first give the definition of state vector. Then the system model in prediction part, observation model and state estimation are explained in sequence.

A. State Vector

State space in our work is 3D space built on the tennis court. State vector X_k contains 3D position factor P_k and 3D velocity factor S_k of tennis ball at discrete time k and gives the current physical motion detail of it. In a form

$$X_k = [P_k, S_k]^T, \quad k \in N \tag{1}$$

Position factor P_k at time k consists of three dimensional factors of the target tennis ball position in 3D space. It is written by real number inside the tennis court area representing the center of gravity position of the target

$$P_k = (x_k, y_k, z_k), \quad k \in N \tag{2}$$

Velocity factor S_k at time k is also written by three dimensional factors representing 3D velocity of the target tennis ball in real number.

$$S_k = (Sx_k, Sy_k, Sz_k), \quad k \in N \tag{3}$$

After defining the state vector of the target tennis ball, we initialize it manually by first several state steps information.

B. System Model

Time evolution of the state vector has been given in a system model with a mixture system noise, which is used for predicting the position and velocity in 3D space of the target tennis ball in next state step. System model is written as

$$\begin{bmatrix} P_k \\ S_k \end{bmatrix} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} P_{k-1} \\ S_{k-1} \end{bmatrix} + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} W_k \tag{4}$$

Where W_k is the current system noise. And T is the time interval between two state steps.

In consideration of changing probability of tennis ball motion in next state step, we design a mixture system noise consists of two components to predict the near future course of ball. One component is the general change noise $W_{k_{general}}$, the other component is the adaptive abrupt change noise $W_{k_{abrupt}}$. The mixture ratio of these two components is decided by the abrupt motion probability. The detail of these two components and the mixture ratio will be explained in part B of section Ⅲ.

C. Observation Model

Multiple visual features of tennis ball and anti-occlusion observation method are used to obtain the likelihood [4].

First, the position and radius in 3D space of current particle are projects the to four image planes according to project principle and tennis ball physical size. Then the likeness of single direction is calculated by the combination of three visual features, color feature, moving feature and edge feature. The region of interest of each particle is decided by projected center coordinate and radius. First, the color feature is computed by the Bhattacharyya Distance between the HSV histograms of target and references. Second, the moving feature is obtained by the background subtraction to remove the complex background influence. The last one is edge feature which measures the circle shape degree of the boundary.

The anti-occlusion observation method is done after getting the likeness of four directions of current particle. It ignores the lowest likeliness and only calculates the higher ones of this particle. This method can eliminate the low likelihood caused by occlusion problem happened in one camera and the result likelihood is appropriate to tracking the target.

D. State Estimation

The weight of every particle is computed after all the particles gaining their likelihood value. The particles state probability is

$$p(X_K|I_k) \approx \sum_{i=1}^{I} w_k^i \, \delta(X_K - X_k^i) \tag{5}$$

Where I_k is the observation space which is a collection of image frames at discrete time k, $\delta(\cdot)$ is the Dirac delta function, I is the total number of particles, w_k^i is the normalized importance weight.

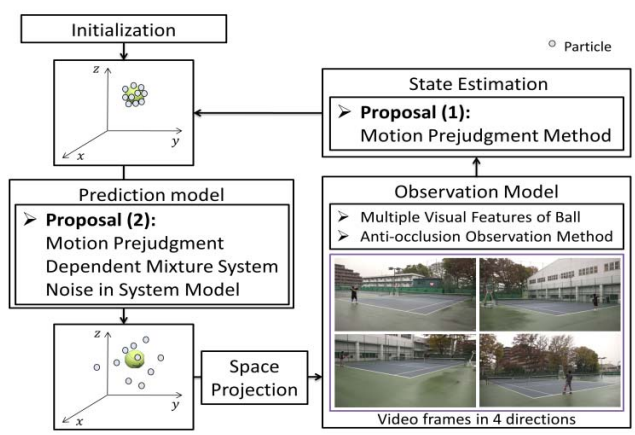

Figure. 1. Whole structure of 3D ball tracking with proposals.

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III. PROPOSAL

To tackle with the high speed and abrupt motion change problems, the mixture system noise dependent on motion prejudgment result is proposed in prediction part of tennis ball tracking process. In this section, firstly, the definition of tennis ball motion and the motion prejudgment method are explained. Secondly, the proposed mixture system noise is put forward with adaptive abrupt motion probability, adaptive abrupt change noise and general change noise which are dependent on the motion prejudgment result.

A. Motion Prejudgment Method of Tennis Ball

The motion of tennis ball is divided into two parts according to whether it is acted by any abrupt external force or not. One part is general motion. It is defined by the ball isn't acted by any abrupt external force, which means it only follows the physical law caused by stable external force, such as gravity and air drag. The other part is abrupt motion. It is determined by the ball gaining an abrupt external force, such as the force caused by player's hit and the force that reacted when hitting the net and court ground.

In addition, the abrupt motion of tennis ball can be separated into three components, smash, bounce and hit the net, according to the different sources of abrupt external force. Firstly, "smash" represents an abrupt motion when tennis ball being hit by players. Secondly, "bounce" is an abrupt motion that the tennis ball hits the ground. Thirdly, "hit the net" is an abrupt motion that tennis ball hitting the net and blocked by it.

The motion prejudgment method is executed at the end of state estimation in tracking process. This method is designed to prejudge the motion of tennis ball in next state step based on current state of ball and players. The result of prejudgment is among the above defined general motion and three abrupt motions along with an abrupt motion probability.

Firstly, for judging whether it is high probability to be smash in next state step , we refer to the following conditions

$$\begin{cases} 1.\ distance\ (ball\ , player) < threshold(1) \\ 2.\ \Delta\ distance\ (ball\ , player) < 0 \end{cases} \tag{6}$$

These two conditions measure the tendency of smash, which is decided by the distances between ball and two players in current state step and the change of them. The 3D positions of players are obtained by tracking result utilizing the method designed by Zhuang's work [10]. In case motion prejudgment is done at the end of the state estimation step, after getting the tracking result of ball and two players of current frame, the distance between them can be calculated. These two formulations compute whether the ball is in the hit scope of one player and at the same time, is flying towards the same player. The threshold(1) is decided by hit scope of players.

Secondly, for calculating the probability of next state motion to be bounce, we estimate these two conditions

$$\begin{cases} 1.\ distance\ (ball\ , ground) < threshold(2) \\ 2.\ \Delta\ distance\ (ball\ , ground) < 0 \end{cases} \tag{7}$$

The first condition indicates that in current state step, the height of ball is low, nearing to hit the ground. The second condition means, considering the current and former height information, the ball is in a falling period and threshold(2) is decided by the average speed of tennis ball.

Thirdly, if next state motion gains a high probability to be hit the net, it needs to satisfy these following three conditions

$$\begin{cases} 1.\ distance(ball, net) < threshold(3) \\ 2.\ \Delta\ distance(ball, net) < 0 \\ 3.\ z_k + Sz_k \times T < threshold(4) \end{cases} \tag{8}$$

The first and second conditions are similar with smash and bounce case expressing that the ball is going closely to the net in current state step. The third condition indicates if the ball keeps on flying, the height of ball will be lower than the net height which leads a high probability that ball hits the net. The threshold(3) is decided by the average speed of tennis ball and threshold(4) is the height of the upper boundary of net.

Finally, if current state doesn't meet any requirements of the above three abrupt motions, it means the prejudged result is general motion. The priority of prejudgment are smash, bounce, hit the net and general motion, which means if current ball state satisfies the first smash conditions, it will judge next state to be smash and not consider other cases' conditions.

B. Motion Prejudgment Dependent Mixture System Noise

The mixture system noise W_k in system model (4) is defined by a formulation as follow

$$W_k = \left(1 - {}^{\alpha}/_{2}\right) \times W_{k_{general}} + {}^{\alpha}/_{2} \times W_{k_{abrupt}} \tag{9}$$

In this equation, α is the probability of one certain abrupt motion. $W_{k_{general}}$ is the general change noise. $W_{k_{abrupt}}$ is the adaptive abrupt change noise.

If the next state motion is general motion, it means the target tennis ball will keep its motion state with a little change according to gravity and air resistance. At this time, we define the noise arousing this kind of little change to be general change noise $W_{k_{general}}$. On the contrary, if the next state motion is one of the abrupt motions, which means the target will gain some abrupt external force, we call current influence noise as abrupt change noise $W_{k_{abrupt}}$. The abrupt motion probability α is the mixture ratio. In summary, in our work, general change noise $\mathrm{W}_{\mathrm{k}_{\mathrm{general}}}$ is invariant and the value of $W_{k_{abrupt}}$ and α is depend on the motion prejudgment result, which means they are adaptive.

Let $\tau_{general}$ follow the Gaussian distribution, suggesting it owns high probability that small change will happen in former value in one dimension.

$$\tau_{general} \sim N(0, {\sigma_1}^2), \sigma_1 = 1.5 \tag{10}$$

Let τ_{abrupt} follow (11), expressing it has high probability to undergo big change from former value in one dimension.

$$\tau_{abrupt} \sim 3.0 - N(0, {\sigma_2}^2), \sigma_2 = 1.0 \tag{11}$$

The general change noise $W_{k_{general}}$ is same in any cases regardless of motion prejudgment result. It contains three dimensions of $\tau_{general}$ and a constant noise A represents average speed of tennis ball, as is shown

$$W_{k_{general}} = \begin{bmatrix} \tau_{general} \\ \tau_{general} \\ \tau_{general} \end{bmatrix} A \sim \begin{bmatrix} N(0, {\sigma_1}^2) \\ N(0, {\sigma_1}^2) \\ N(0, {\sigma_1}^2) \end{bmatrix} A, \sigma_1 = 1.5 \tag{12}$$

The abrupt motion probability α and abrupt change noise $W_{k_{abrupt}}$ are adapted according to different prejudgment result.

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Firstly, if the motion prejudgment result is smash, the future external force mainly comes from the racket of player, indicating that none of the three dimensions of velocity will maintain its former value. In this case,

$$\alpha(smash) = 1 - \frac{distance(ball, player)}{threshold\ (1)} \tag{13}$$

$$W_{k_{abrupt}} = \begin{bmatrix} \tau_{abrupt} \\ \tau_{abrupt} \\ \tau_{abrupt} \end{bmatrix} A \sim \begin{bmatrix} 3.0 - N(0, {\sigma_2}^2) \\ 3.0 - N(0, {\sigma_2}^2) \\ 3.0 - N(0, {\sigma_2}^2) \end{bmatrix} A, \sigma_2 = 1.0 \tag{14}$$

Secondly, in case the motion prejudgment result is bounce, the abrupt change of velocity mainly occurs on z direction.

$$\alpha(bounce) = 1 - \frac{distance\ (ball\ , ground)}{threshold\ (2)} \tag{15}$$

$$W_{k_{abrupt}} = \begin{bmatrix} \tau_{general} \\ \tau_{general} \\ \tau_{abrupt} \end{bmatrix} A \sim \begin{bmatrix} N(0, {\sigma_1}^2) \\ N(0, {\sigma_1}^2) \\ 3.0 - N(0, {\sigma_2}^2) \end{bmatrix} A, \begin{matrix} \sigma_1 = 1.5 \\ \sigma_2 = 1.0 \end{matrix} \tag{16}$$

Thirdly, if hit the net case is the result of motion prejudgment, the velocities in x and y directions tend to shift.

$$\alpha(hit\ the\ net) = 1 - \frac{distance\ (ball\ , net)}{threshold\ (3)} \tag{17}$$

$$W_{k_{abrupt}} = \begin{bmatrix} \tau_{abrupt} \\ \tau_{abrupt} \\ \tau_{general} \end{bmatrix} A \sim \begin{bmatrix} 3.0 - N(0, {\sigma_2}^2) \\ 3.0 - N(0, {\sigma_2}^2) \\ N(0, {\sigma_1}^2) \end{bmatrix} A, \begin{matrix} \sigma_1 = 1.5 \\ \sigma_2 = 1.0 \end{matrix} \tag{18}$$

Finally, on condition that the motion prejudgment result turns out to be general motion, the α equals to zero.

IV. EXPERIMENT

Experiment is based on videos recorded outside by four cameras located at four corners of the tennis court in a cloudy day. The total video sequences number is 13 and contains 3940 frames in all. The resolution of video is 1920×1080 pixels and the frame rate is 60 fps. Shutter speed of the camera is set as 6000 per second for the sake of no motion blur. The videos are recoded in 3rd, December 2015.

To evaluate the performance of our proposal, our tracking result is shown by the green circle drawn in 4 image planes. Based on the drawn figures, we use tracking success rate to show the result which is given in the following. Firstly, if the projected circle exactly covers or covers part of the ball in one direction image plane, we define tracking at this direction is success and this direction is called success direction. Secondly, in case at least 3 success directions at one time out of total 4 are gotten, we confirm that tracking in this frame is success and this frame is called success frame.

$$\text{Tracking success rate} = \frac{\sum \text{success frame}}{\sum \text{frame}} \times 100\% \tag{19}$$

For comparison, we also do experiments on the same sequences by conventional work [4]. The results of our work and conventional work are shown in TABLE I.

TABLE I. EVALUATION RESULT

Sequence number	Proposal work tracking success rate	Conventional work tracking success rate
Sequence-1	344/350 = 98.57 %	135/350 = 38.57 %
Sequence-2	289/371 = 77.90 %	141/371 = 38.01 %
Sequence-3	394/395 = 99.75 %	145/395 = 36.71 %
Sequence-4	135/144 = 93.75 %	144/144 = 100.0 %
Sequence-5	200/200 = 100.0 %	130/200 = 65.00 %
Sequence-6	216/276 = 78.26 %	173/276 = 62.68 %
Sequence-7	270/270 = 100.0 %	270/270 = 100.0 %
Sequence-8	161/210 = 76.67 %	142/210 = 67.62 %
Sequence-9	323/542 = 59.59 %	147/542 = 27.12 %
Sequence-10	140/140 = 100.0 %	135/140 = 96.43 %
Sequence-11	334/469 = 70.91 %	337/469 = 71.86 %
Sequence-12	138/304 = 45.39 %	72/304 = 23.68%
Sequence-13	253/269 = 94.05 %	137/269 = 50.93 %
Total	3197/3940 = 81.14 %	2108/3940 = 53.50 %

Fig. 2 shows the tracked 3D position result of sequence-1, sequence-3, sequence-5 and sequence-7 drawn by MATLAB. The coordinate indexing value is centimeter. Fig. 3 shows the 350×200 pixels part of images out of 80-th frame and 160-th frame in sequence-1 from four directions to explain the tracking result in image frames. In these figures, the green circle means the tracking result of tennis ball and the blue rectangles represent the two players.

As a summary of experiment result, firstly, our proposal achieves 81.14% tracking success rate and three sequences reach 100.0%. Secondly, since the conventional work gets 53.50% success rate, it proves that our proposal gains 27.64% improvement. However, on consideration of the tracking failure cases, the severe occlusion problem caused by players and other objects happened in more than one camera still needs to be solved, which happens in sequence-2, sequence-6, sequence-8, sequence-11 and sequence-12. What's more, the limitation of videos recording environment, the cloudy weather, makes the computation of visual features of ball inaccurate and leads the tracking result unstable, such as result of sequence-4 becomes worse and sequence-9 remains not ideal. On the whole, although there still remaining some points to be improved, our proposal performs well and gains a considerable improvement for tennis ball 3D position tracking.

V. CONCLUSION

In this paper, aiming at achieving high success rate of tracking tennis ball in 3D space for tennis game analysis, we have proposed a motion prejudgment dependent mixture system noise in system model by particle filter to solve the prediction problem caused by high speed and abrupt motion change of the target tennis ball. The system model is written by time evolution of former state vector and system noise. The system noise is a mixture of general change noise and adaptive abrupt change noise. The mixture ratio and certain abrupt

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motion are obtained by the motion prejudgment result of ball in next state step. The motion prejudgment method is based on current state of ball and players. Experiments on 13 videos recording a tennis game outside in cloudy day by 4 cameras measure the performance of our proposal. The results are shown by green circle in 4 directions image and the tracking success rate is calculated to be 81.14%. According to the experiment result, we receive 27.64%improvement compared with conventional work, proving our proposal works well.

As for future work, the observation model still needs to be modified to overcome the severe occlusion problem. Moreover, once tracking failure happens, the position hardly recovers. However, comparing with other technologies, our work can be extended to other topics for tennis game analysis, such as event detection, automatic score judgment and tactics analysis in case our work can refer to both position and velocity in 3D space of target tennis ball.

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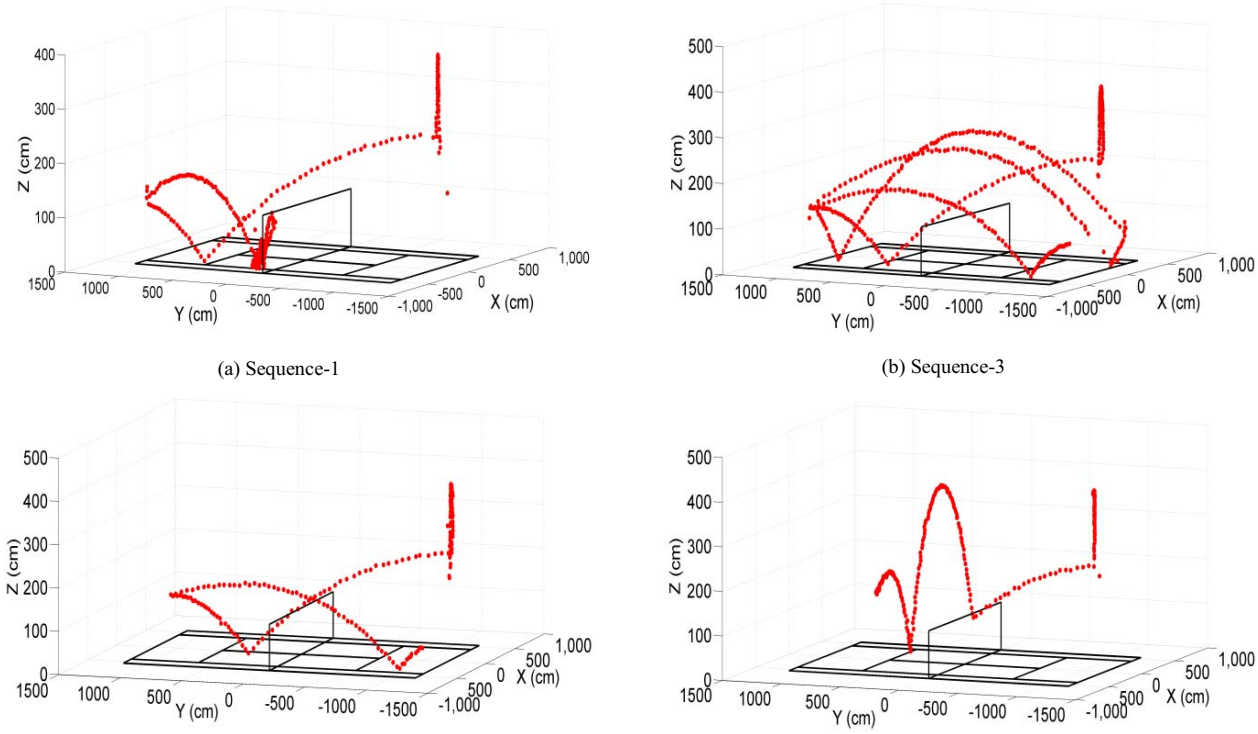

(a) Sequence-1

(b) Sequence-3

(c) Sequence-5

(d) Sequence-7

Figure. 2. Experiment result in 3D space

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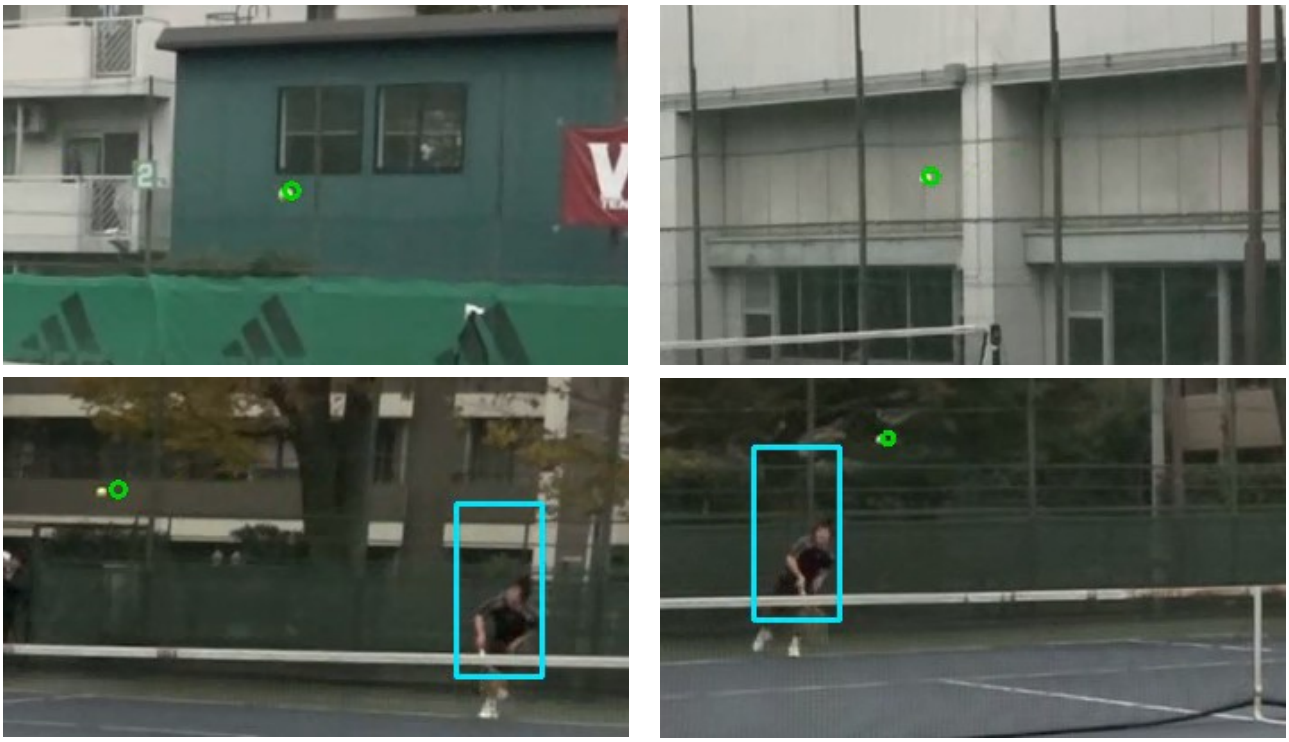
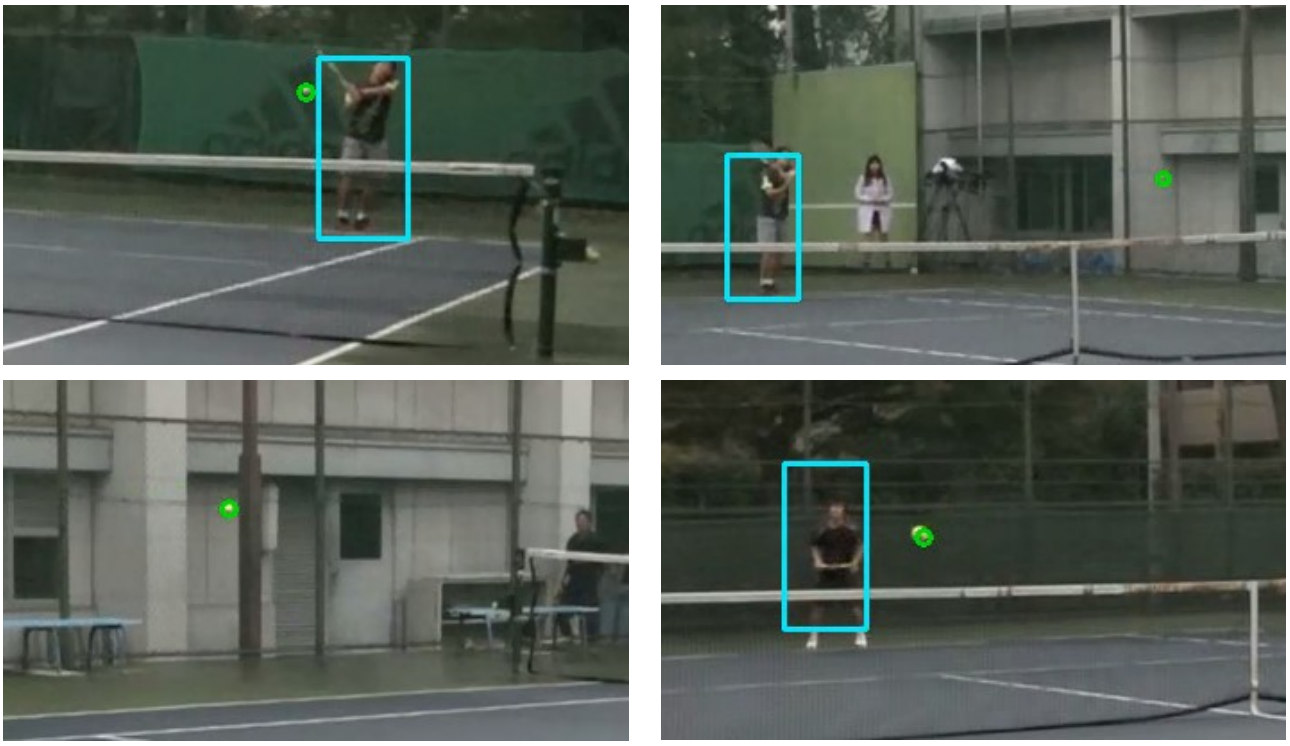


80-th frame



160-th frame

Figure. 3. Experiment result by 4 cameras image of sequence-1

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